Bashir Agro Admin Panel
Phase 5: Task Assignment Management Module

Phase Scope

Implement ONLY Task Assignment Management after Phase 4.

Included:
- Task assignment database
- CRUD
- Farmer selection
- Task template selection
- Farm location selection
- Scheduling
- Priority
- Status management
- Search
- Filters
- Pagination
- Activity logs
- Tests

Not included:
- Flutter app
- REST API
- Task submission
- Photo evidence
- GPS validation
- Geofence checking
- Notifications

Objective

Connect the completed modules:

Farmers (Phase 2)
+
Farm Locations (Phase 3)
+
Task Templates (Phase 4)

into a complete assignment workflow.

Database Table: task_assignments

Fields:

id

task_template_id
farmer_id
farm_location_id

scheduled_date

start_time
end_time

priority

status

assigned_by
assigned_at

started_at
completed_at

created_at
updated_at
deleted_at

Field Rules

task_template_id:
References task_templates.

farmer_id:
References farmers.

farm_location_id:
References farm_locations.

priority:
normal, important, urgent.

status:
assigned, in_progress, completed, cancelled.

Initial status:
assigned.

Use soft delete.

Relationships

Farmer:
hasMany TaskAssignments.

TaskTemplate:
hasMany TaskAssignments.

FarmLocation:
hasMany TaskAssignments.

TaskAssignment belongsTo:

- Farmer
- TaskTemplate
- FarmLocation
- User (assigned_by).

Model

Create:

app/Models/TaskAssignment.php

Requirements:

- Use SoftDeletes
- Add fillable fields
- Add relationships

Relationships:

farmer()
taskTemplate()
location()
assigner()

Migration

Create task_assignments migration.

Foreign keys:

task_template_id -> task_templates.id

farmer_id -> farmers.id

farm_location_id -> farm_locations.id

assigned_by -> users.id

Use proper constraints to protect historical data.

Controller

Create:

app/Http/Controllers/Admin/
TaskAssignmentController.php

Methods:

index()
create()
store()
show()
edit()

update()
destroy()

Use Eloquent and keep controllers clean.

Form Requests

Create:

StoreTaskAssignmentRequest.php

UpdateTaskAssignmentRequest.php

Validate:

farmer_id
task_template_id
farm_location_id
scheduled_date
start_time
end_time
priority
status

Ensure end time is after start time.

Assignment Creation Workflow

Admin:

Select Farmer
↓
Select Task Template
↓
Select Location
↓
Choose Date and Time
↓
Choose Priority
↓
Save Assignment

Backend:

Validate
↓
Check relationships
↓
Create assignment
↓
Create activity log
↓
Redirect

Business Rules

Rules:

1. One assignment belongs to exactly one farmer.

2. One farmer can have many assignments.

3. One assignment belongs to one task template.

4. One assignment belongs to one location.

5. Prevent accidental duplicate assignments.

Assignment List Page

Create:

resources/views/assignments/index.blade.php

Display:

- Farmer
- Task
- Location
- Date
- Time
- Priority
- Status
- Assigned By
- Actions

Remove hard-coded assignment data.

Search

Implement server-side search.

Example:

/admin/assignments?search=rahim

Search:

- Farmer name
- Task name
- Location name

Filters

Implement:

Status:
- All

- Assigned
- In Progress
- Completed
- Cancelled

Priority:
- Normal
- Important
- Urgent

Date filters:
- Today
- This week
- Custom date

Pagination

Use Laravel pagination.

Recommended:

20 assignments per page.

Maintain filters using:

withQueryString().

UI Requirements

Preserve Bashir Agro design:

- Bengali labels
- Green theme
- Existing cards
- Existing forms
- Existing tables
- Responsive layout

Do not redesign.

Dashboard Integration

Replace static task counts.

Example:

```
TaskAssignment::whereDate(
'scheduled_date',
today()
)->count();
```

Dashboard must use PostgreSQL data.

Activity Logging

Create logs:

assignment.created
assignment.updated
assignment.deleted
assignment.status_changed

Store:

user_id
action
subject_type
subject_id
description
old_values
new_values
ip_address

Routes

Add:

```
Route::resource(
'assignments',
TaskAssignmentController::class
);
```

Under:

/admin/assignments

Security

Use:

- Authentication middleware
- CSRF
- Form Requests
- Authorization
- Mass assignment protection
- Server-side validation

Never trust browser validation only.

Testing

Create:

tests/Feature/
TaskAssignmentManagementTest.php

Test:

- Authentication
- Create assignment
- Relationships
- Validation
- Duplicate prevention
- Update
- Soft delete
- Search
- Filters
- Pagination

Antigravity Prompt

```
Proceed to PHASE 5 ONLY: Task Assignment Management Module.

Use the existing Bashir Agro Laravel project after Phase 4.

Do not redesign the UI.

Do not implement:
- Flutter app
- REST API
- Task submission
- Photo upload
- GPS validation
- Geofence checking
- Notifications

Implement:

- task_assignments table
- TaskAssignment model
- Relationships
- Controller CRUD
- Form Requests
- Farmer selection
- Task selection
- Location selection
- Scheduling
- Priority
- Status
- Search
- Filters
- Pagination
- Soft delete
- Activity logging
- Feature tests

Preserve the existing Bengali green agricultural UI.

Remove hard-coded assignment data from JavaScript.

Database must be the source of truth.
```

```
Run migrations.
Run php artisan test.
Run vendor/bin/pint.

Report:
- files created
- files modified
- database changes
- routes added
- test results

STOP after Phase 5.
Do not start Phase 6 automatically.
```